

Jinho Cha, Ph.D.

Lawrenceville, Georgia, United States • jcha@gwinnetttech.edu • ORCID

RESEARCH PROFILE

Machine Learning • Large-Scale Data Analytics • Statistical Modeling • AI Evaluation • Decision Intelligence • Optimization

Applied AI and data researcher with a strong quantitative foundation in statistical inference, stochastic modeling, machine learning, and optimization. Proven record of converting complex, high-stakes analytical problems into validated data-driven decision systems and leading funded R&D programs.

TECHNICAL SKILLS

- **Core:** Python, R, pandas, NumPy, scikit-learn, LightGBM
- **Applied Experience:** PyTorch, TensorFlow, XGBoost, Java, Jupyter
- **Familiarity:** SQL, AWS (S3, EC2, SageMaker)
- **Methods:** Predictive Modeling, Statistical Inference, Count Models, Calibration, Clustering, NLP/Text Mining, Simulation, Robust Optimization, Decision Analytics

SELECTED RESEARCH HIGHLIGHTS

- **Machine Learning & Predictive-Prescriptive Analytics** — Built reservation-level cancellation models on **119,390 hotel bookings** and integrated calibrated risk estimates into rolling booking control, improving mean realized profit by **33.22%**. Published in *Computers & Industrial Engineering*, 2026.
- **Statistical Risk Modeling & Robust Optimization** — Developed Negative Binomial models for clustered, overdispersed risk and extended them to distributionally robust sequential control. Empirical analyses used S&P 500 jump-count data and demonstrated tail-risk protection under model uncertainty. Two *JIMO* articles published in 2026.
- **AI Decision-System Orchestration** — Developed a stochastic dynamic optimization framework for switching among heterogeneous analytical modules with different quality, latency, reliability, and transition costs. Established stability-aware switching and hysteresis results. Published in *JIMO*, 2026.
- **Mechanism Design & Decentralized Resource Allocation** — Developed a smart-contract-mediated allocation framework with equilibrium, convergence, fairness, and shock-resilience guarantees for decentralized decision systems. Published in *JIMO*, 2026.
- **Simulation & Empirical Decision Analytics** — Built a 10-year stochastic health-policy simulation calibrated with MEPS and NHANES data to evaluate adherence, intervention timing, cost, and ROI. Published in *BMC Public Health*, 2025.

PROFESSIONAL EXPERIENCE

COMPUTER SCIENCE DIVISION, GWINNETT TECHNICAL COLLEGE

Faculty, Computer Programming

LAWRENCEVILLE, GA

2025 – Present

Conduct AI/data research while teaching programming and computational problem solving.

KOREA RESEARCH INSTITUTE FOR DEFENSE TECHNOLOGY PLANNING AND ADVANCEMENT (KRIT)

SOUTH KOREA

Senior Researcher

2021 – 2025

Led data-intensive R&D spanning analytics, simulation, intelligent systems, logistics, and technology decision support.

SUNGKYUNKWAN UNIVERSITY, GRADUATE SCHOOL OF ADVANCED DEFENSE

SEOUL, SOUTH KOREA

Lecturer, Artificial Intelligence (Concurrent Appointment)

During KRIT tenure

Delivered graduate-level instruction in applied artificial intelligence.

DEPARTMENT OF MATHEMATICS, KOREA MILITARY ACADEMY

SEOUL, SOUTH KOREA

Assistant Professor

2017 – 2021

Conducted quantitative research and taught probability, statistics, linear algebra, and optimization.

REPUBLIC OF KOREA ARMY HEADQUARTERS / JOINT CHIEFS OF STAFF

SOUTH KOREA

Operations Research & Analytics Lead

Prior to 2017

Applied simulation, operational data analysis, and quantitative decision methods to large-scale mission-critical planning.

FUNDED R&D LEADERSHIP

- **\$1.0M Project Manager** — Led a national R&D program from problem definition and analytical design through validation and implementation.
- **\$750K Co-PI** — Co-led an intelligent coastal-surveillance program integrating sensing, analytics, AI learning models, and decision support.
- **\$450K Co-PM / \$200K Co-PI / \$100K PI** — Additional leadership in simulation, training analytics, personnel decision systems, and intelligent technologies.

EDUCATION & HONORS

Ph.D., Industrial Engineering, Clemson University • **M.S., Industrial Engineering, University of Florida** • **M.S., Industrial Engineering, Texas A&M University** • **B.S., Electronic Engineering, Korea Military Academy**

Presidential Commendation, Republic of Korea (2025) • **Ministerial Commendation, Ministry of National Defense (2023)** • **Army Chief of Staff Commendation (2019)**